Reza Ghorbani Saber

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Profile

- Operations Research specialist with expertise in combinatorial optimization, scheduling, exact/decomposition algorithms, metaheuristics, and stochastic optimization.
- C/C++ developer experienced in implementing optimization algorithms, custom data structures, solver integrations, and computationally intensive scheduling applications.
- Experienced in translating industrial planning and scheduling problems into mathematical and algorithmic solutions.
- Experienced with Gurobi, IBM ILOG CPLEX, CP Optimizer, Python, Git, CMake, HPC, and Docker.

Professional Experiences

Ghent University & Flanders Make, Ghent, Belgium — Feb 2022-Present
PhD Researcher & Algorithm Developer

- Architected and implemented high-performance C++ scheduling engines for human-centric Flexible Job Shop Scheduling (FJSP).
- Formulated both deterministic and stochastic optimization models to address real-world operational uncertainties, workforce side constraints, and dual-resource machine/operator allocation.
- Implemented exact decomposition schemes (Logic-Based Benders Decomposition, MIP, CP) and local search metaheuristics to improve solution quality and cut runtime on large-scale benchmarks.
- Developed and implemented algorithmic development within the collaborative industry project S4FMS, building modular C++ and Python APIs to integrate optimization engines into partner architectures and industrial testbeds.
- Interfaced production-grade mathematical solvers (Gurobi, IBM ILOG CPLEX, CP Optimizer) with custom C++ data structures and graph-based search routines.

Taha Ghaleb Toos company (TGT Co), Mashhad, Iran — May 2020- Oct 2021
Project Planning and Control Specialist

- Directed production planning and resource scheduling across specialized manufacturing facilities (Machine, CNC, Laser Cladding, and Assembly shops) for over 20 industrial R&D projects
- Developed an automated database and reporting pipeline linking MS Access and Excel, delivering real-time shop-floor tracking, capacity visibility, and financial progress dashboards to executive management.
- Mitigated operational bottlenecks and contractor schedules by translating physical machine constraints and job priorities into actionable production timetables

Education

Ghent University, Ghent, Belgium — 2022-Present
Doctor of philosophy in Industrial Engineering and Operations Research

Ferdowsi University of Mashhad, Mashhad, Iran — 2016-2018
MSc in Industrial Engineering-Systems Optimization

Ferdowsi University of Mashhad, Mashhad, Iran — 2012-2016
BSc in Industrial Engineering

Publications

Ghorbani Saber, R., Leyman, P., & Aghezzaf, E. H. (2026). Decision support for the integrated flexible job shop scheduling and operator timetabling problem. *International Journal of Production Research*, 1-28. https://doi.org/10.1080/00207543.2026.2642881

Ghorbani Saber, R., & Ranjbar, M. (2022). Minimizing the total tardiness and the total carbon emissions in the permutation flow shop scheduling problem. *Computers & Operations Research*, *138*, 105604. https://doi.org/10.1016/j.cor.2021.105604

Ranjbar, M., & **Ghorbani Saber, R**. (2021). A variable neighborhood search algorithm for transshipment scheduling of multi products at a single station. *Applied Soft Computing*, *98*, 106736. https://doi.org/10.1016/j.asoc.2020.106736

Languages

English	Full professional proficiency (C1, IELTS 7.5)
Dutch	Intermediate (B1)
Persian	Native